RISHABH JAIN

| **Email**: j_rishabh@outlook.com | **Address**: Dublin, Ireland | **Website:** www.rishabhja.in
| **Scholar:** https://scholar.google.com/citations?user=z8K3q8cAAAAJ | **LinkedIn**: www.linkedin.com/in/rishabhjain16 |
| **GitHub**: www.github.com/rishabhjain16 | **HuggingFace**: www.huggingface.co/rishabhjain16 |

DATA SCIENTIST | AI ENGINEER | ML ENGINEER | RESEARCH SCIENTIST

I am a Data Scientist, AI Engineer, and ML Engineer with a PhD in AI for Speech from the University of Galway, Ireland. My work focuses on visual speech recognition, multimodal audio-visual AI, speech-to-text, text-to-speech, and large language model-based speech technologies. I have experience developing, evaluating, and benchmarking AI systems that combine audio, visual, and language information to improve robustness, generalization, and human-aligned performance. I am especially interested in building practical AI systems that perform reliably in real-world conditions.

STRENGTHS AND TECHNICAL EXPERTISE

Soft skills:
Teamwork, communication, leadership, Research-driven thinking, problem-solving, ethical AI advocacy, and responsible AI.

Languages and ML Libraries:
SQL, R, Python, C, C++, C#, Java, HTML, CSS, NodeJS, Pandas, NumPy, SciPy, Librosa, Transformers, Sklearn, Seaborn, Tensorflow and PyTorch.

Softwares and Cloud Platforms:
Anaconda, Spyder, Jupyter Notebook, VS Code, Eclipse, AWS, GCP,Colab, Docker, Git, Bitbucket, Ollama, Claude Code, GitHub Copilot, Linux.

PROFESSIONAL EXPERIENCE

Trinity College Dublin, Ireland — **November 2024 - Present**
Research And Teaching Fellow (Post Doctoral Researcher)

- Working on the Speech Space project to advance **multimodal audio-video speech understanding**, with a focus on robust, human-aligned models that combine visual and acoustic speech cues.
- Developing standardized pipeline for **generalized AVSR** and **lipreading** evaluation across benchmarks, enabling consistent data processing and reproducible experiments.
- Developing and evaluating **LLM-based audio-video speech systems** that integrate visual and acoustic cues for robust speech understanding, with attention to accuracy, reliability, and responsible AI use..
- Exploring evaluation methods that better reflect **human perceptual behavior**, with attention to transparency, fairness, and the limits of benchmark performance.

Trinity College Dublin, Ireland — **January 2026 - May 2026**
Teaching Fellow

- Designed and delivered an **Advanced AI module** with **lectures**, **labs**, **assignments**, and **assessments** across NLP, computer vision, speech and audio, and efficient fine-tuning.
- Led hands-on notebooks and lab sessions connecting theory to implementation across **transformer architectures**, tokenisation, attention mechanisms, **speech-to-text** and **text-to-speech** systems, multimodal models, and parameter-efficient adaptation, including **BERT**, **GPT**, **Vision Transformers**, **CLIP**, **Whisper**, **LoRA**, and **QLoRA**.
- Supervised student projects across **reinforcement learning**, **generative AI**, **multimodal systems**, **speech** and accessibility, vision, and financial machine learning.

Oxford Wave Research, Oxford, UK — **February 2024 - May 2024**
Research Scientist (Contract)

- Conduct research on training the models in the domain of **Speech-To-Text** and **LLMs** for low resource languages.
- Working with engineers to train and deploy models for production, and optimizing models for real-world use cases such as improving ASR for low resource **Albanian speech**, which led to improved transcription accuracy by **20%**.
- Proficient in finetuning, and deploying advanced LLMs (eg. Llama, Gemma, Gemini, Command-R, Mistral). Leveraged diverse LLM tools and frameworks such as **Ollama, Langchain, LM Studio and llama.cpp**.
- Creating a pipeline combining **Whisper ASR, speaker diarization, speaker segmentation and LLM summarization**, enhancing model capabilities for nuanced tasks in NLP and NLU for law and forensics use cases, cutting down manual summarisation time by 40%, thereby boosting productivity for legal and forensic professionals.
- Experienced in **LLM Quantization** methodologies, implementing efficient model compression techniques to optimize inference speed and resource utilization while maintaining model fidelity.
- Conducted evaluations of LLMs across multiple linguistic domains, leading to **precise optimizations** that improved model performance, accuracy, and applicability in diverse, real-world tasks critical to legal and forensic analysis.
- Managed and analyzed sensitive **forensic speech data**, including law enforcement and criminal data, ensuring responsible and ethical AI deployment in forensic investigations.
- Developed models and methodologies to detect and mitigate **deep fakes**, ensuring the integrity and reliability of AI applications in forensic and legal contexts.

Xperi Galway (now part of Tobii), Ireland **October 2020 - January 2024**
Researcher (Consultant)

- Designed and deployed **edge AI models (STT and TTS)** in collaboration with Xperi, significantly enhancing child-user experience by enabling seamless real-time speech interaction in the DAVID smart-toy.
- Collaborated with cross-functional teams, including engineers, linguists, and product managers from Xperi Ireland and Xperi USA for **optimising AI models** and **deployment** in production environment.
- Developed new methodologies for ethical and responsible **data collection** and **annotation** of **child speech** (automatic labelling), addressing challenges like data scarcity and noise, improving the robustness of child speech ASR and TTS models.
- Developed **synthetic child speech datasets**, resulting in a 5-10% increase in the accuracy of **child ASR systems**.
- Implemented **data anonymisation techniques** for child speech datasets, enhancing privacy protections and ensuring compliance with stringent ethical standards and regulations.

University of Galway, Ireland **October 2020 - January 2024**
Research Assistant

- Project: DTIF-DAVID, Aim: Development of a **multimodal (sound and vision)** AI processing platform with **low cost** and **low powe**r consumption to be used for the creation of voice-enabled toys.
- Conduct a comprehensive review of existing research and literature related to **child speech technologies (ASR/TTS**).
- Developed methodologies for **ASR** and **TTS** with child speech, from training to evaluation of models, resulting in synthetic speech datasets that upheld privacy standards and **GDPR** standards, which also led to the creation of open-source resources and speaker recognition models tailored for various accents in English, spoken by children.
- Developing applications for **speech data collection**, creating methodologies for **data annotations** and **cleaning**.
- Attending conferences, workshops, seminars, and knowledge-sharing activities within research community.
- Documentations of research, preparing reports/presentations and contributing towards **novel publications**.
- Setting up and managing distributed training pipelines (leveraging multiple GPUs), coordinating data preprocessing and augmentation, monitoring training progress, and evaluating model performance using appropriate metrics.
- Led the adaptation of **Whisper** models to better recognize child speech, achieving between **20% - 50% improvement** in accuracy over different child speech datasets.
- Collaborated with Data Protection Officers (DPOs) to ensure the **ethical collection** and use of child speech data.
- Helped shape **data collection** and **compliance practices** to ensure privacy-conscious, responsible handling, and risk-aware use of child speech data.

EDUCATION

PhD in Artificial Intelligence for Speech, University of Galway, Ireland **September 2021 - May 2024**
Thesis: Child speech understanding and generation via neural ASR and TTS models

MSc in Data Analytics, University of Galway, Ireland **September 2019 - August 2020**
Grade: 1.1 Honours
Thesis: Semi automatic toolkit for analysis of video, audio and text data based on face recognition in videos.

B.Tech in Computer Science and Engineering with spec. in Bioinformatics, VIT University, India **July 2015- May 2019**
CGPA: 8.73 (equivalent to 1.1 Honours)

PUBLICATION AND CONFERENCES

Publications:

- **R. Jain** and N. Harte, "From Hype to Insight: Rethinking Large Language Model Integration in Visual Speech Recognition," **ICASSP 2026** , Barcelona, Spain, 2026, pp. 18717-18721, doi: 10.1109/ICASSP55912.2026.11464703.
- **R. Jain,** A. Barcovschi, M. Y. Yiwere, D. Bigioi, P. Corcoran and H. Cucu, "A WAV2VEC2-Based Experimental Study on Self-Supervised Learning Methods to Improve Child Speech Recognition," in **IEEE Access,** vol. 11, pp. 46938-46948, 2023, doi: 10.1109/ACCESS.2023.3275106.
- **Jain, R**., Barcovschi, A., Yiwere, M., Corcoran, P., Cucu, H. (2023) Adaptation of Whisper models to child speech recognition. Proc. **INTERSPEECH 2023**, 5242-5246, doi: 10.21437/Interspeech.2023-935

Conferences:

- Presented at Interspeech 2023, ICASSP 2026, and IEEE Region 8 SYP Congress (Tunis, Tunisia) as Chair, representing the University of Galway Student Branch.
- Organised a special session on Child Speech Technologies at the SpeD 2023 Conference, Bucharest, Romania
- Served as a reviewer for Interspeech, ICASSP, ASRU, Computer Speech & Language, IEEE Access, IMVIP, and SpeD.

REFERENCES

- Dr. Anil Alexander, CEO, Oxford Wave Research, Oxford, UK, anil@oxfordwaveresearch.com
- Dr. Peter Corcoran, IEEE fellow, Professor at the University of Galway, Lead PI in DAVID, HELIUS and d-real projects, peter.corcoran@universityofgalway.ie
- Dr. Gabriel Costache, VP of Machine Learning at Fotonation, gabriel.costache@fotonation.com